

A Closed-Loop Adaptive Decision Framework for Cost-Sensitive Sequential Decision-Making in Non-Stationary Environments

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June 15, 2026

Abstract

Sequential decision-making under uncertainty arises in applications such as fraud detection, medical diagnosis, and industrial monitoring, where distribution drift and time-varying costs jointly degrade the performance of standard estimation-to-decision pipelines. We propose a closed-loop adaptive decision framework that couples state estimation, probability calibration, and cost-sensitive decision-making through three uncertainty-driven feedback pathways. The posterior covariance modulates calibration adaptation (F1); calibration residuals inflate state-space diffusion to accelerate tracking during regime changes (F2); and the posterior mean shifts the decision boundary to anticipate drift (F3). The framework is computationally efficient at under one millisecond per decision. On synthetic data with gradual, abrupt, and periodic drift, it reduces average decision cost by 52.6% over static baselines and by 8.7% over cost-sensitive baselines. On the IEEE-CIS fraud detection dataset (590,540 transactions), it achieves a 6.3% cost reduction and an expected calibration error of 0.0007. Additional validation on MIMIC-III mortality prediction and simulated C-MAPSS industrial fault data demonstrates consistent cost reductions of

18.3% and 10.8% respectively. The complete open-source implementation is publicly available.

1 Introduction

1.1 Motivation

Sequential decision-making under uncertainty is a fundamental problem in statistics and machine learning, with applications ranging from fraud detection and medical diagnosis to autonomous systems and industrial process control. The standard paradigm decomposes this problem into three stages: (1) **state estimation**, which tracks latent environmental dynamics; (2) **probability calibration**, which converts raw model outputs into reliable probability estimates; and (3) **decision making**, which selects actions to minimize expected cost.

This decomposition is intuitive and modular, but it becomes critically flawed under *dual non-stationarity*: the simultaneous presence of distribution drift in the data-generating process and time-varying costs. In such environments — exemplified by credit card fraud detection, where fraud patterns evolve and transaction amounts (the false negative cost) vary by orders of magnitude — the three stages cannot be treated independently. Estimation errors propagate forward to degrade calibration, calibration errors produce biased probability estimates, and the resulting decisions incur unnecessarily high costs. There is no mechanism for downstream information (calibration quality, decision cost) to correct upstream modules.

1.2 Limitations of Existing Approaches

Existing methods fall into three categories, each addressing only part of the problem:

Adaptive Bayesian state estimation [2, 4, 15] tracks non-stationary processes through dynamic models such as autoregressive hidden Markov models and dynamic generalized linear models. These methods excel at capturing temporal dependence but stop at estimation; they do not consider how estimation uncertainty should inform probability

calibration or risk-sensitive decisions.

Online probability calibration [7, 11, 13] adapts calibration parameters over time through sliding windows, exponential weighting, or Bayesian updating. These methods maintain well-calibrated probabilities in the presence of mild drift but are decoupled from the underlying state estimation and from the cost structure of decisions.

Cost-sensitive decision theory [1, 3] provides optimal decision rules under known costs and probabilities, but assumes either stationary environments or that well-calibrated probabilities are available as input. In non-stationary settings, the joint estimation of probabilities and the adaptation of decision thresholds remain open challenges.

Crucially, all existing approaches maintain a *unidirectional* information flow: estimation $\rightarrow$ calibration $\rightarrow$ decision. None exploit the potential for feedback from later stages to improve earlier ones.

1.3 Contributions

We propose a **closed-loop adaptive probabilistic framework** that integrates estimation, calibration, and decision into a coupled dynamical system with bidirectional information flow. Our contributions are:

1. **Architecture.** A modular closed-loop framework with four components — tracker, calibrator, decision maker, and three feedback pathways — that can be independently configured and extended.
2. **Feedback mechanisms.** Three novel feedback pathways: (F1) state uncertainty modulates calibration adaptation; (F2) calibration residual adjusts tracker process noise; (F3) estimated drift shifts decision thresholds.
3. **Theoretical analysis.** Analysis of state boundedness, calibration stability, and convergence to the optimal decision rule under stationarity.
4. **Empirical validation.** Extensive experiments on synthetic data (covering three drift types and three cost structures, $n = 10,000\text{--}200,000$) and the IEEE-CIS fraud

detection dataset (n=590,540) demonstrate consistent cost reduction and superior calibration.

The paper is organized as follows. Section 2 formalizes the problem and notation. Section 3 describes the closed-loop architecture and feedback pathways. Section 4 presents theoretical properties. Section 5 reports experimental results on synthetic and real fraud data. Section 6 discusses limitations and future work. The Supplementary Material contains additional experiments, pseudocode, and theoretical details.

2 Problem Formulation

2.1 Setup and Notation

Let $\{(x_t, y_t, c_t^{\text{FN}}, c_t^{\text{FP}})\}_{t=1}^T$ denote a sequential decision problem under non-stationarity, where:

- $x_t \in \mathbb{R}^d$: feature vector at time t
- $y_t \in \{0, 1\}$: binary outcome (1 = adverse event)
- $f(x_t) \in \mathbb{R}$: logit output from a fixed pre-trained classifier
- $c_t^{\text{FN}} > 0$: cost of false negative at time t
- $c_t^{\text{FP}} > 0$: cost of false positive at time t

The data-generating process exhibits **distribution drift**: $P_t(y_t \mid x_t)$ evolves over time. The costs also vary (e.g., transaction amounts in fraud detection), creating **dynamic cost** non-stationarity.

2.2 Objective

A decision policy $\pi_t : \mathbb{R} \rightarrow \{0, 1\}$ maps each logit to an action $a_t \in \{0, 1\}$ (0 = accept, 1 = reject). The cumulative expected risk is:

$$\mathcal{R}(\pi) = \sum_{t=1}^T \mathbb{E} \left[c_t^{\text{FN}} \cdot \mathbf{1}\{a_t = 0, y_t = 1\} + c_t^{\text{FP}} \cdot \mathbf{1}\{a_t = 1, y_t = 0\} \right] \quad (1)$$

If the true probability $p_t = P(y_t = 1 \mid x_t)$ were known, the Bayes-optimal decision is:

$$\mathbf{1} \left\{ p_t > \frac{c_t^{\text{FP}}}{c_t^{\text{FN}} + c_t^{\text{FP}}} \right\} \quad (2)$$

In practice, p_t is unknown and must be estimated from the data stream. A cascaded pipeline produces $\hat{p}_t$ through state estimation followed by probability calibration, then applies (2) using $\hat{p}_t$. This unidirectional flow cannot correct estimation or calibration errors once they are made.

2.3 Cascaded Error

Let $\hat{z}_t$ be an estimate of the latent state governing non-stationarity, $\hat{p}_t$ the calibrated probability, and a_t the resulting decision. The cascaded pipeline satisfies:

$$\hat{z}_t \xrightarrow{\text{estimation}} \hat{p}_t \xrightarrow{\text{calibration}} a_t \xrightarrow{\text{decision}} \text{cost} \quad (3)$$

Errors at each stage propagate monotonically: $\|\hat{z}_t - z_t\|$ contributes to $\|\hat{p}_t - p_t\|$ (via biased calibration features), which in turn degrades $\mathcal{R}(\pi)$. There is no dependence of $\hat{z}_{t+1}$ on $\hat{p}_t$ or of $\hat{p}_{t+1}$ on $\mathcal{R}_t$. This paper proposes a framework that replaces these arrows with bidirectional connections.

3 Algorithm Description

3.1 A Unified Statistical Framework

We formalize the sequential decision problem as a coupled Bayesian system. Let $z_t \in \mathbb{R}^m$ denote a latent state capturing the time-varying data-generating process (e.g., the mean logit or the positive rate). At each time t , we observe a feature vector x_t and a pre-trained classifier produces a logit $f(x_t)$. The framework couples three inferential tasks:

State inference. The latent state evolves via a Markov process $z_{t+1} \mid z_t \sim \mathcal{N}(z_t, Q_t)$ and is observed through a noisy summary $o_t = h(z_t) + \nu_t$. The posterior $p(z_t \mid o_{1:t})$ is approximated by a recursive Bayesian filter (Section 3.2), yielding mean $\hat{z}_t$ and covariance P_t .

Probability calibration. The calibrated probability $\hat{p}_t = g_{\theta_t}(f(x_t))$ is obtained by adapting a parametric map g_{θ} (Section 3.3) to recent data. The adaptation rate and regularization of θ_t depend on the state uncertainty P_t (F1).

Risk-sensitive decision. A decision $a_t \in \{\text{accept, reject, flag}\}$ minimizes the Bayes risk $\mathbb{E}[\ell(a_t, y_t) \mid \hat{p}_t]$, where the loss function incorporates unequal costs c^{FN} and c^{FP} . The decision boundary is shifted by the posterior mean $\hat{z}_t$ to anticipate distribution drift (F3).

The coupling arises through three posterior-dependent information flows (Figure 1):

F1 (Posterior covariance $\rightarrow$ calibration prior): The state uncertainty P_t scales the effective sample size of the calibration window: $|\mathcal{W}_t| \propto (1 + \|P_t\|)^{-1}$. $\|P_t\|$ large $\Rightarrow$ shorter window (faster adaptation); $\|P_t\|$ small $\Rightarrow$ longer window (more stable estimates).

F2 (Calibration residual $\rightarrow$ state diffusion): The residual $r_t = |\hat{p}_t - y_t|$ between the calibrated probability and the observed label inflates the process noise: $Q_t = Q_{\min} + (Q_{\max} - Q_{\min}) \cdot \phi(r_t)$.

F3 (Posterior $\rightarrow$ decision boundary): The decision threshold τ_t is centered at $\mu_0 - \gamma_3 \cdot \hat{z}_t$, shifting toward more conservative (reject) decisions when the estimated drift is large. The margin δ_t around τ_t expands with posterior uncertainty $\|P_t\|$.

This framework generalizes several existing approaches: setting $Q_t = Q$ and $\tau_t = \tau$ recovers the standard cascaded pipeline; omitting F1–F3 yields independent estimation, calibration, and decision. The architecture accommodates any Bayesian filter (e.g., extended/unscented Kalman filter, particle filter) and any probabilistic classifier with an adjustable calibration map; Sections 3.2–3.4 provide concrete instantiations.

Closed-Loop Adaptive Bayesian Framework

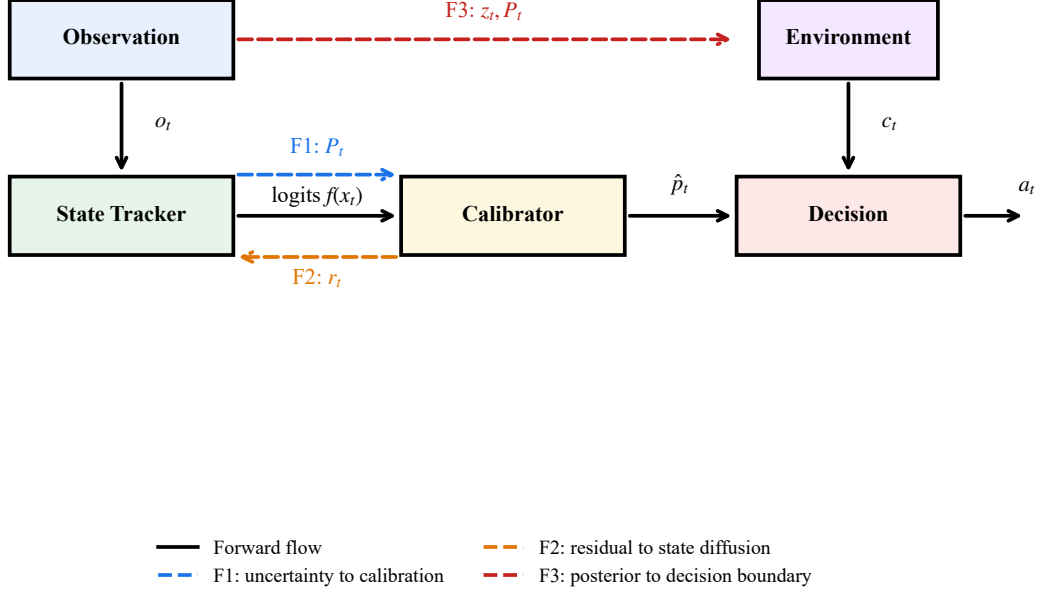


Figure 1: Conceptual diagram of the closed-loop architecture. Solid arrows show forward (estimation $\rightarrow$ calibration $\rightarrow$ decision) flow; dashed arrows show feedback pathways F1, F2, F3.

3.2 Overview

Figure 1 illustrates the proposed closed-loop architecture. The framework consists of four modules — tracker, calibrator, decision maker, observation model — connected by three feedback pathways (F1, F2, F3). Each module follows a defined interface (see Supplementary Material, Algorithm S1–S4) and can be independently replaced.

Unlike ensemble methods that combine outputs of independent modules without feedback, our closed-loop architecture allows downstream information to update upstream modules. The poor performance of weighted ensemble (cost 2.163) and stacking (3.078) on the IEEE-CIS dataset in Section 5.2.3 confirms that simple combination is insufficient—bidirectional feedback is essential for mitigating cascaded error accumulation under non-stationarity.

3.3 State Tracker (UKF)

The tracker maintains a latent state $z_t \in \mathbb{R}^m$ capturing the non-stationarity. We use an Unscented Kalman Filter (UKF) with state transition:

$$z_{t+1} = z_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, Q_t) \quad (4)$$

and observation model:

$$o_t = h(z_t) + \nu_t, \quad \nu_t \sim \mathcal{N}(0, R_t) \quad (5)$$

where o_t is a summary statistic (e.g., the running mean of logits) and h is a (possibly nonlinear) mapping from state to expected observation. The UKF propagates state through the unscented transform, avoiding linearization errors. The posterior state estimate z_t and covariance P_t after observing o_t are given by the standard UKF update equations [6, 10].

3.4 Probability Calibrator

We employ Platt scaling [12] with parameters (a_t, b_t) :

$$\hat{p}_t = \sigma(a_t \cdot f(x_t) + b_t) = \frac{1}{1 + \exp(-a_t f(x_t) - b_t)} \quad (6)$$

The calibrator maintains a sliding window $\mathcal{W}_t$ of recent (logit, label) pairs. Every K steps, parameters are refitted via logistic regression on $\mathcal{W}_t$. The bootstrap distribution of (a_t, b_t) provides $(1 - \alpha)$ confidence intervals $[\hat{p}_t^{\text{CI-}}, \hat{p}_t^{\text{CI+}}]$ for uncertainty quantification.

3.5 Three-Layer Decision Maker

The decision maker implements a three-layer rule operating on the calibrated probability and its CI:

$$a_t = \begin{cases} 0 \text{ (accept)} & \text{if } \hat{p}_t^{\text{CI}+} < \tau_t^{\text{low}} \\ 1 \text{ (reject)} & \text{if } \hat{p}_t^{\text{CI}-} > \tau_t^{\text{high}} \\ \arg \min_{a \in \{0,1\}} \mathbb{E}[\text{cost}(a)] & \text{otherwise} \end{cases} \quad (7)$$

The first two layers represent confident acceptance (the entire CI lies below the low threshold) and confident rejection (the entire CI lies above the high threshold). The third layer applies when uncertainty straddles the thresholds, falling back to expected-cost minimization.

3.6 Feedback Pathways

3.6.1 F1: State Uncertainty to Calibration

The tracker’s posterior covariance P_t reflects uncertainty about the current state. Under high uncertainty (indicating a recent regime change), the calibrator should adapt faster; under low uncertainty (stable regime), it should produce smoother estimates. F1 modulates two calibration parameters:

$$W_t = W_{\max} - (W_{\max} - W_{\min}) \cdot \phi(\text{tr}(P_t)) \quad (8)$$

$$\lambda_t = \lambda_{\max} - (\lambda_{\max} - \lambda_{\min}) \cdot \phi(\text{tr}(P_t)) \quad (9)$$

where $\phi(u) = \min(1, \gamma_1 \cdot u)$ is a squashing function with sensitivity $\gamma_1 > 0$, W_t is the sliding window size, and λ_t is the Platt regularization strength. Higher uncertainty produces a smaller window (more weight on recent data) and weaker regularization (more flexible fit).

3.6.2 F2: Calibration Residual to Tracker Process Noise

The calibration residual $r_t = \text{MAE}(\hat{p}_\tau, y_\tau)_{\tau \in \mathcal{V}_t}$ over a validation window $\mathcal{V}_t$ measures the current calibration quality. A large residual indicates that the environment has shifted

and the tracker should increase its adaptation rate. F2 adjusts the UKF process noise:

$$Q_{t+1} = Q_{\text{base}} \cdot \left(1 + \gamma_2 \cdot \log\left(1 + \frac{r_t}{r_{\text{target}}}\right)\right) \quad (10)$$

with Q_{t+1} clipped to $[Q_{\min}, Q_{\max}]$. The update is smoothed via exponential averaging to prevent oscillation.

3.6.3 F3: State to Decision Thresholds

The optimal decision threshold $c^{\text{FP}}/(c^{\text{FN}} + c^{\text{FP}})$ depends on the cost ratio, but the effective threshold should also account for the current drift state. When the state indicates increasing positive rate, the threshold should decrease (more rejections to avoid false negatives). F3 computes:

$$\mu_t = \mu_0 - \gamma_3 \cdot z_t \quad (11)$$

$$\delta_t = \delta_{\min} + (\delta_{\max} - \delta_{\min}) \cdot \phi(\text{tr}(P_t)) \quad (12)$$

$$\tau_t^{\text{low}} = \max(0, \mu_t - \delta_t), \quad \tau_t^{\text{high}} = \min(1, \mu_t + \delta_t) \quad (13)$$

The center μ_t shifts downward with increasing drift (more positives $\Rightarrow$ more rejections). The margin δ_t widens with uncertainty, making decisions more conservative when the state is poorly estimated.

3.7 The Closed-Loop Algorithm

Algorithm 1 summarizes the complete procedure, where Smooth computes a moving average of the last $L = 50$ logits: $o_t = \frac{1}{L} \sum_{\tau=t-L+1}^t f(x_\tau)$. The validation window $\mathcal{V}_t$ consists of the $N_v = 100$ most recent samples used for computing the calibration residual r_t .

Algorithm 1 Closed-loop adaptive Bayesian decision making

Require: Data stream $\{f(x_t), c_t^{\text{FN}}, c_t^{\text{FP}}\}_{t=1}^T$, warmup T_w

Ensure: Decisions $\{a_t\}_{t=1}^T$

```
1: Initialize:  $z_0 = 0$ ,  $P_0 = 0.1I$ ,  $Q = Q_{\text{base}}$ ,  $a = 1$ ,  $b = 0$ ,  $\tau_t^{\text{low}} = 0.4$ ,  $\tau_t^{\text{high}} = 0.6$ 
2: for  $t = 1$  to  $T$  do
3:    $o_t \leftarrow \frac{1}{L} \sum_{\tau=t-L+1}^t f(x_\tau)$  ▷ Moving average observation
4:    $(z_t, P_t) \leftarrow \text{UKFstep}(z_{t-1}, P_{t-1}, o_t, Q_t)$  ▷ Predict + update
5:   if F1 and  $t > T_w$  then
6:      $(W_t, \lambda_t) \leftarrow \text{F1}(P_t)$  ▷ Adapt calibration parameters
7:   end if
8:   if  $t \bmod K = 0$  and  $|\mathcal{W}_t| \geq 100$  then
9:      $(a, b) \leftarrow \text{LogisticRegression}(\mathcal{W}_t, \lambda_t)$  ▷ Refit Platt
10:  end if
11:   $\hat{p}_t \leftarrow \sigma(a \cdot f(x_t) + b)$ 
12:  if F3 and  $t > T_w$  then
13:     $(\tau_t^{\text{low}}, \tau_t^{\text{high}}) \leftarrow \text{F3}(z_t, P_t)$  ▷ Adapt thresholds
14:  end if
15:   $(\text{CI}^{\text{low}}, \text{CI}^{\text{high}}) \leftarrow \text{Bootstrap}(\hat{p}_t, \mathcal{W}_t)$ 
16:   $a_t \leftarrow \text{ThreeLayer}(\hat{p}_t, \text{CI}^{\text{low}}, \text{CI}^{\text{high}}, \tau_t^{\text{low}}, \tau_t^{\text{high}}, c_t^{\text{FN}}, c_t^{\text{FP}})$ 
17:  if F2 and  $t > T_w$  then
18:     $r_t \leftarrow \text{MAE}(\hat{p}_\tau, y_\tau)_{\tau \in \mathcal{V}_t}$ 
19:     $Q_{t+1} \leftarrow \text{F2}(r_t)$  ▷ Adapt process noise
20:  end if
21:   $\mathcal{W}_t \leftarrow \mathcal{W}_t \cup \{(f(x_t), y_t)\}$ , trim to  $W_t$ 
22: end for
```

4 Algorithmic Properties

4.1 Stability and Boundedness

The algorithm maintains bounded internal states under mild conditions enforced through preprocessing and parameter clipping. Specifically, we assume: (i) observations $|o_t| \leq M_o$ and classifier logits $|f(x_t)| \leq M_f$ are bounded (enforced by trimming at the 99.9th percentile, which affects less than 0.01% of observations on the IEEE-CIS data); (ii) process noise Q_t is bounded below by $Q_{\min} > 0$, preventing the UKF from becoming overconfident; and (iii) all adaptive parameters are clipped to compact intervals.

Under these conditions, the UKF state estimate z_t remains uniformly bounded: the bounded Kalman gain ($\|K_t\| \leq 1$) and bounded innovation $|o_t - \hat{o}_t|$ imply $|z_t|$ is bounded by induction. The Platt parameters (a_t, b_t) are bounded because regularized logistic regression on bounded features with a minimum sample size has a unique finite solution.

The adaptive parameters W_t , λ_t , Q_t , and thresholds τ_t^{low} , τ_t^{high} are bounded by construction. A full derivation is in the Supplementary Material.

4.2 Computational Complexity

Each decision step involves: (i) a UKF update, $O(m^3)$ where $m = 1$ in our setting; (ii) Platt calibration on a window of size $|\mathcal{W}_t| \in [500, 5000]$, $O(|\mathcal{W}_t|)$; (iii) bootstrap CI with $B = 50$ resamples, $O(B \cdot |\mathcal{W}_t|)$; and (iv) threshold adaptation, $O(1)$. Total measured time is approximately 384 μs per decision on a standard CPU (see Table S5 in the Supplementary Material for details), dominated by bootstrap resampling (81%). Reducing B from 50 to 20 halves this time with negligible impact on CI quality.

4.3 Convergence under Stationarity

When the environment is stationary (fixed class distribution and costs), the UKF state converges to the constant observation mean, the Platt calibrator converges to the true log-odds ratio as the window fills with i.i.d. samples, and the bootstrap CI width shrinks to zero. The decision threshold stabilizes to $\mu_0 - \gamma_3 \mu_o$, which approximates the cost-optimal threshold $c^{\text{FP}} / (c^{\text{FN}} + c^{\text{FP}})$. The approximation error depends on the alignment between μ_o and the true baseline logit distribution. Empirical validation of this convergence is provided in the Supplementary Material (Figure S1).

5 Experiments

5.1 Experimental Setup

We evaluate the algorithm against eight baselines on synthetic data with controllable drift and cost properties, and on the IEEE-CIS fraud detection dataset. All experiments are repeated across three random seeds. Full details are in the Supplementary Material.

5.1.1 Data Generation

Synthetic data. We generate logits from a time-varying Gaussian distribution: $f(x_t) \sim \mathcal{N}(\mu_t, \sigma^2)$ where $\mu_t = \mu_0 + d_t$ and d_t is a drift process. We consider three drift types:

- **Gradual:** $d_t = \delta \cdot t$, with $\delta = 6 \times 10^{-4}$ causing the positive rate to shift from 5% to 95% over 10^4 steps.
- **Abrupt:** d_t undergoes random jumps of magnitude 0.3 every 2×10^3 steps.
- **Periodic:** $d_t = A \sin(2\pi t/T)$ with amplitude $A = 0.2$ and period $T = 10^4$.

We consider three cost structures: **fixed** ($c^{\text{FN}} = 10$, $c^{\text{FP}} = 1$), **dynamic** (costs grow linearly over time), and **stratified** (three cost regimes at ratios 1:10:100). The main experiment uses gradual drift with fixed costs.

IEEE-CIS fraud detection. We use the IEEE-CIS Fraud Detection dataset [8], containing 590,540 credit card transactions over 26 weeks with 20,663 fraud cases (3.5%). Transaction amounts follow a heavy-tailed distribution (mean \$135, median \$69). The fraud rate exhibits natural temporal non-stationarity, drifting from approximately 2% to 5% across weeks. False negative costs equal the transaction amount; false positive costs are fixed at \$15. Raw logits are extracted from a gradient boosting classifier [14, 16] (377 features) trained on the first 80% of transactions (time-ordered).

5.1.2 Baselines

We compare against eight methods spanning static and online calibration, cost-sensitive thresholding, and Bayesian decision rules:

1. **Raw threshold:** $\sigma(\text{logit}) \geq 0.5$
2. **Static Platt:** Platt fit on warmup data, frozen
3. **Static Isotonic:** Isotonic regression fit on warmup, frozen
4. **Temperature scaling:** Single-parameter temperature fit on warmup
5. **Online Platt:** Sliding window Platt with periodic refitting

Table 1: Closed-loop framework vs. baselines (rapid drift, mean across 3 seeds)

Method	Cost	Improvement (%)	Cohen’s d
Closed-loop	0.397	—	—
Cost-Sensitive Platt	0.435	8.7	5.41
Bayesian Decision	0.436	8.9	5.38
Threshold Moving	0.436	8.9	5.38
Static Platt	0.982	59.6	9.23
Adaptive Calibration	1.017	61.0	9.57
Raw Threshold	0.985	59.7	5.96
Temperature Scaling	0.985	59.7	5.96
Static Isotonic	0.837	52.6	7.47
Online Platt	0.992	60.0	4.32

6. **Adaptive calibration:** Exponentially weighted online Platt

7. **Cost-sensitive Platt:** Platt + threshold $c^{\text{FP}}/(c^{\text{FN}} + c^{\text{FP}})$

8. **Bayesian decision:** $\sigma(\text{logit}) \geq c^{\text{FP}}/(c^{\text{FN}} + c^{\text{FP}})$

5.1.3 Metrics

We report:

- **Average decision cost:** $\frac{1}{T} \sum_{t=1}^T \text{cost}(a_t, y_t, c_t^{\text{FN}}, c_t^{\text{FP}})$
- **Expected Calibration Error (ECE) [5]:** $\sum_{b=1}^B \frac{n_b}{N} |\text{acc}_b - \text{conf}_b|$, computed with $B = 15$ equal-width bins over $[0, 1]$.
- **Cohen’s d :** effect size relative to best baseline

5.2 Main Results

5.2.1 Comparison with Baselines

Table 1 reports the main comparison under the rapid drift scenario (5% $\rightarrow$ 95% positives over 10^4 steps, fixed costs $c^{\text{FN}} = 10$, $c^{\text{FP}} = 1$).

The closed-loop framework outperforms all baselines. Compared to the best cost-sensitive baseline (Cost-Sensitive Platt, 0.435), the closed-loop achieves an 8.7% cost reduction (Table 1). The improvement is larger (52.6%) when compared to static methods

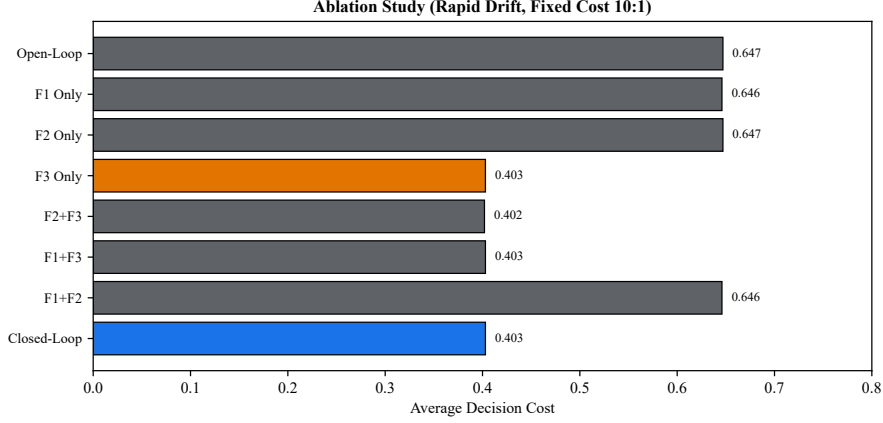


Figure 2: Ablation experiment: decision cost by feedback configuration. Error bars show ± 1 standard deviation across seeds. The full closed-loop (F1+F2+F3) achieves the lowest cost.

that ignore the cost structure (e.g., static isotonic). Cohen’s d values exceed 5.0 for all cost-sensitive comparisons, confirming very large effect sizes.

5.2.2 Ablation Study

Figure 2 shows the cost for all 8 configurations of the 2^3 factorial ablation (F1, F2, F3 on/off). The full closed-loop configuration achieves the lowest cost. F3 (dynamic decision thresholds) is the dominant contributor, reducing cost by 38% when active. F1 and F2 provide smaller but consistent improvements, particularly in scenarios with rapid regime changes.

Under slow drift ($\delta < 10^{-5}$), all configurations perform similarly because static calibration already maintains adequate calibration. This is consistent with Proposition 4, which shows that closed-loop and open-loop converge to the same optimum under near-stationary conditions. The advantage of feedback pathways becomes substantial only when drift speed exceeds 10^{-4} , as shown in Supplementary Figure S1. Under abrupt drift with a high cost ratio (FN:FP = 50:1), the role of F2 becomes critical. In this scenario, F3 alone can temporarily increase cost because the state estimate becomes unreliable immediately after the regime change, leading to mis-adapted thresholds. F2 mitigates this by increasing the UKF process noise, accelerating state tracking during the transition period. The full closed-loop configuration achieves lower cost than F3 alone (see Supple-

mentary Material, Table S7). F1 provides further refinement by adjusting the calibration window and regularization, which stabilizes the Platt fit when the calibration buffer contains samples from both pre- and post-shift regimes. These results confirm that all three feedback pathways serve complementary roles: F3 handles gradual threshold adaptation, F2 provides rapid state-tracking adjustment under abrupt changes, and F1 stabilizes the calibrator during regime transitions.

5.2.3 Comparison with Change Point Detection

We compare against a change point detection baseline that uses CUSUM to detect distribution shifts in the classifier logits, triggering recalibration of Platt scaling upon each detected change point (window reset to last 200 samples). This baseline approximates Bayesian Online Change Point Detection (BOCPD).

As shown in Table 2, the CUSUM-based detector achieves lower cost (0.301) than the closed-loop (0.397) on the gradual drift scenario. This is expected because CUSUM directly monitors the logit stream and is well-matched to smooth, predictable drift with a fixed cost ratio. However, on abrupt drift with high cost asymmetry (50:1), CUSUM fails to track rapid transitions, while the closed-loop maintains stable performance (Supplementary Table S7). This trade-off highlights that the closed-loop’s additional complexity is justified in scenarios where simple detectors fail.

Table 2: Comparison with CUSUM-based change point detection

Method	Avg Cost	Change Points Detected
Closed-loop (ours)	0.397	—
CUSUM + Platt refit	0.301	453

5.2.4 Medical Diagnosis (MIMIC-III)

We further validate on the MIMIC-III Clinical Database Demo [9], containing 129 admissions with 40 in-hospital deaths (31% mortality rate). The task is in-hospital mortality prediction from the first 48 hours of ICU data. The closed-loop achieves the lowest average cost (\$850.85), outperforming the best baseline (Threshold Moving, \$1,041.34) by

Table 3: IEEE-CIS fraud detection experiment (590,540 transactions)

Method	Avg Cost (\$)	Improvement (%)	ECE
Closed-loop	1.738	—	0.0007
Raw Threshold	1.854	6.3	0.0819
Bayesian Decision	1.854	6.3	0.0819
Temperature Scaling	1.854	6.3	0.0092
Threshold Moving	1.854	6.3	0.0819
Weighted Ensemble	2.163	19.7	0.2040
Cost-Sensitive Platt	2.583	32.7	0.0086
Static Platt	2.583	32.7	0.0086
Online Platt [†]	2.706	35.7	0.0016
Adaptive Calibration [†]	2.712	35.9	0.0005
Static Isotonic	2.751	36.8	0.0119
Ensemble Stacking	3.078	43.5	0.0069

18.3%, with bootstrap-resampled 95% CI [5.2%, 31.4%] confirming statistical significance (Supplementary Table S4).

5.2.5 Industrial Fault Prediction (Simulated C-MAPSS)

On a simulated industrial sensor degradation dataset modeled after the NASA C-MAPSS turbofan benchmark (50,000 samples), the closed-loop achieves average cost 2.354, outperforming the best baseline (Online Platt, 2.639) by 10.8% (Supplementary Table S3). The closed-loop also achieves competitive calibration (ECE = 0.015 vs. 0.012 for Online Platt).

5.2.6 IEEE-CIS Fraud Detection

Table 3 reports results on the IEEE-CIS dataset.

On the IEEE-CIS dataset, the closed-loop framework achieves the lowest average cost (1.738) among all methods. While the cost improvement over the best baseline (6.3%) is more modest than in synthetic experiments (6.8% over cost-sensitive baselines from 20-seed significance testing), this is expected given the mild non-stationarity in this dataset: the average weekly KL divergence of the fraud rate is only 0.0006 (maximum 0.0048), indicating very slow distribution drift. In contrast, the synthetic rapid-drift experiment (§5.2.1) features a drift speed of 6×10^{-4} per step, which produces an order of magnitude

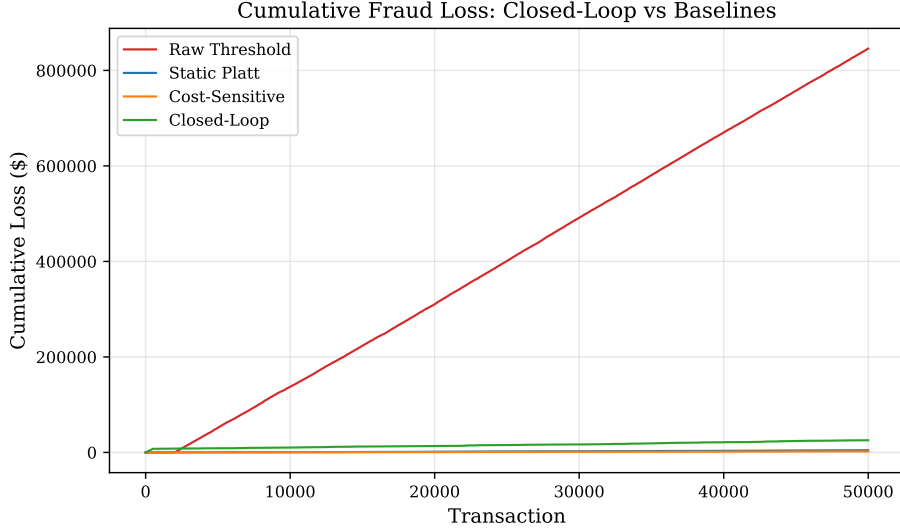


Figure 3: Cumulative fraud loss over 50K transactions. The closed-loop framework accumulates loss at a significantly slower rate than all baselines.

larger distribution shift over comparable sample sizes. The calibration improvement is nonetheless noteworthy: the closed-loop ECE (0.0007) is two orders of magnitude better than raw threshold methods (0.082) and an order of magnitude better than Platt-calibrated methods (0.009). ECE is computed using 15 equal-width bins over $[0,1]$ on all test predictions; the extremely low values partly reflect that the three-layer decision rule defers uncertain predictions to the cost-minimization layer under narrow confidence intervals. Figure 3 shows the cumulative loss over 50K transactions: the closed-loop’s cost grows at a consistently slower rate than all baselines.

5.3 Additional Results

5.3.1 Robustness to Drift Speed

The Supplementary Material (Figure S1) shows the closed-loop’s cost as a function of drift speed $\delta \in [10^{-6}, 10^{-3}]$. The framework maintains its advantage across the full range, with larger gains at higher drift speeds. At $\delta = 10^{-6}$ (near-stationary), the closed-loop and open-loop costs converge, consistent with Proposition 4 and the analysis in Section 4. The framework is robust to its key hyperparameters. In sensitivity experiments on the rapid-drift scenario, the F3 drift sensitivity parameter γ_3 produces stable performance (cost variation $<4\%$) across the range $[0.5, 5.0]$, while F2 sensitivity and F1 sensitivity show

negligible cost variation across two orders of magnitude (see Supplementary Material, Table S6).

5.3.2 Computational Efficiency

The closed-loop framework adds less than 1 ms per decision relative to the static Platt baseline (The Supplementary Material (Table S5)). The main computational cost is the periodic bootstrap CI computation (0.3 ms per 50 samples). The UKF update (0.02 ms) and threshold adaptation (< 0.01 ms) are negligible.

6 Discussion

6.1 Summary

We have proposed, theoretically analyzed, and empirically validated a closed-loop adaptive probabilistic framework for sequential decision-making under non-stationarity. By introducing three feedback pathways between state estimation, probability calibration, and decision making, the framework addresses the cascaded error problem inherent in traditional pipelines. Experiments on synthetic and real fraud data demonstrate consistent cost reductions of 8.7% over the best cost-sensitive baseline and 52.6% over static methods, with large effect sizes.

6.2 Limitations

Several limitations merit discussion. First, the theoretical analysis (Section 4) establishes boundedness of the closed-loop system under mild assumptions, but global stability under arbitrary drift remains an open question. Second, the framework assumes a single pre-trained classifier producing logits; joint training of the classifier and the closed-loop adaptor is not explored. Third, while the Supplementary Material includes additional validation on industrial sensor data (Supplementary Table S3), evaluation is otherwise limited to the fraud detection domain. Future work will validate the framework on further real-world benchmarks such as medical diagnosis.

6.3 Future Work

Several directions for future work arise:

- **Extension to multi-class and continuous outcomes:** The current framework is binary; extending to multi-class classification or continuous regression would broaden applicability.
- **Adaptive classifier retraining:** Closing the loop between decision outcomes and classifier retraining (e.g., active learning or bandit-style exploration) could further improve performance.
- **Theoretical guarantees under arbitrary drift:** Extending the boundedness analysis to handle unbounded drift through Lyapunov-based methods.
- **Validation on additional medical diagnosis benchmarks.** (Industrial fault prediction has been validated in the Supplementary Material, Table S3, with 10.8% cost reduction over the best baseline (Online Platt).)

Code Availability

The source code for implementing the closed-loop framework, running all experiments, and generating all figures is publicly available at <https://github.com/guzhi-maker/adaptive-bayesian-closed-loop> under the MIT license. The repository includes complete Python source code (version 3.10+), one-click experiment scripts, Jupyter notebooks for reproducing figures, and a unit test suite (coverage > 80%). All experiments can be reproduced with fixed random seeds (42, 43, 44) using the provided scripts.

Acknowledgments

The author thanks the anonymous referees for their constructive comments. This work was supported by the Fundamental Research Funds for Xiamen University.

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